GROUP 5 / MATH 4175

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Cryptography Project 2

Vigenere cipher: coincidence analysis, keyword recovery, and decryption.

Part 1 - Checking the key length

The ciphertext has **1,260 letters**. For a candidate length m, each substring takes every m-th letter, starting at positions 1 through m. These are interleaved subsequences, not consecutive chunks.

```
IC = sum f(f - 1) / [n(n - 1)]
```

Here f is a letter's count and n is the substring length. Every individual IC is listed below; averages are only a supplement. The course benchmarks are English approximately 0.065 and uniform random 1/26 approximately 0.03846.

m = 6

String	n	IC
y1	210	0.042287537
y2	210	0.042606516
y3	210	0.040555935
y4	210	0.040555935
y5	210	0.042059695
y6	210	0.042014126

m = 7

String	n	IC
y1	180	0.071260087
y2	180	0.071260087
y3	180	0.068156425
y4	180	0.061390441
y5	180	0.068342644
y6	180	0.062259466
y7	180	0.067908132

m = 8

String	n	IC
y1	158	0.040312828
y2	158	0.039829074
y3	158	0.043699105
y4	158	0.043295977
y5	157	0.041727911
y6	157	0.044177691
y7	157	0.040503021
y8	157	0.037889923

Key length	Mean IC	Mean |IC - 0.065|
6	0.041679957	0.023320043
7	0.067225326	0.004039638
8	0.041429441	0.023570559

Yes, the results support m = 7. Its seven ICs range from 0.061390441 to 0.071260087, close to 0.065. In contrast, the m = 6 and m = 8 values cluster near 0.04, closer to a uniform-random distribution. Length 7 has the smallest mean absolute distance from 0.065.

With the correct period, each substring uses one Caesar shift, preserving its letter frequencies and IC. An incorrect period mixes different shifts. This supports 7 among the tested candidates, rather than proving the exact period by itself.

Part 2 - Recovering the keyword

For each of the seven substrings and each candidate shift g = 0, ..., 25, compute the dot product between English letter probabilities and the cyclically shifted observed frequencies:

```
M_g = sum p_i * f_((i + g) mod 26) / n
```

Entries below are **100 * M_g** (percentage scale), as in the course table. Thus 6.62 represents M_g = 0.0662. The largest score in each substring column is highlighted.

g	y1	y2	y3	y4	y5	y6	y7
0	3.31	3.54	3.76	4.25	**6.72**	3.89	3.86
1	3.61	3.13	4.05	3.03	3.91	4.60	3.23
2	3.97	3.38	4.68	3.50	3.52	3.55	3.92
3	4.93	4.32	3.89	4.12	3.01	3.67	4.32
4	4.05	4.44	4.24	**6.28**	4.07	3.19	3.62
5	4.15	3.78	3.84	3.56	2.86	3.52	4.02
6	3.83	4.23	4.40	3.19	3.80	3.58	4.11
7	4.91	4.23	3.76	3.52	3.94	3.35	4.71
8	3.79	4.08	3.34	4.65	3.18	4.02	4.07
9	3.39	3.80	3.42	3.42	3.49	3.19	3.64
10	3.05	3.33	3.88	3.37	4.10	3.83	2.98
11	4.06	3.08	3.41	3.70	4.83	4.15	3.52
12	3.06	3.73	3.28	3.39	3.91	**6.37**	4.04
13	3.36	3.61	4.53	4.09	4.29	3.83	3.53
14	4.07	3.65	3.54	3.86	3.84	3.50	4.12
15	3.55	4.41	3.14	4.28	4.25	3.14	3.37
16	3.69	3.44	3.70	3.33	3.81	4.25	3.04
17	4.25	2.84	**6.66**	4.40	3.38	3.45	3.50
18	**6.62**	4.02	3.93	3.96	3.39	3.65	**6.56**
19	3.75	**6.83**	3.22	4.61	3.63	3.47	4.27
20	3.34	4.17	2.96	3.66	3.53	3.12	3.14
21	3.27	3.11	4.10	3.63	3.35	3.96	2.89
22	4.00	3.37	3.18	2.94	4.76	4.00	5.09
23	2.98	4.51	3.71	3.51	3.51	4.77	3.50
24	3.50	3.31	3.81	4.21	3.24	3.62	3.55
25	3.62	3.77	3.68	3.65	3.79	4.42	3.50

```
Winning shifts: 18, 19, 17, 4, 0, 12, 18
```

Using A = 0, ..., Z = 25 gives **S, T, R, E, A, M, S**. The recovered keyword is **STREAMS**, a meaningful English word. Its decryption yields the coherent English text in Part 3.

Part 3 - Decrypted plaintext

Keyword: **STREAMS**. Decrypt by subtracting the repeating key modulo 26. Spaces, punctuation, and sentence capitalization below were added for readability without changing any decrypted letters.

The Department of Justice has been and will always be committed to protecting the liberty and security of those whom we serve.

In recent months, however, we have on a new scale seen mainstream products and services designed in a way that gives users sole control over access to their data. As a result, law enforcement is sometimes unable to recover the content of electronic communications from the technology provider, even in response to a court order or duly authorized warrant issued by a federal judge.

For example, many communications services now encrypt certain communications by default, with the key necessary to decrypt the communications solely in the hands of the end user. This applies both when the data is in motion over electronic networks or at rest on an electronic device.

If the communications provider is served with a warrant seeking those communications, the provider cannot provide the data because it has designed the technology such that it cannot be accessed by any third party.

We do not have any silver bullets, and the discussions within the executive branch are still ongoing. While there has not yet been a decision whether to seek legislation, we must work with Congress, industry, academics, privacy groups, and others to craft an approach that addresses all of the multiple competing concerns that have been the focus of so much debate.

But we can all agree that we will need ongoing, honest, and informed public debate about how best to protect liberty and security in both our laws and our technology.

Verification: Removing spaces and punctuation gives exactly 1,260 letters. Re-encrypting them with STREAMS reproduces every letter of the original ciphertext.

Reference: MATH 4175, Section 2.4, slides 13-18 and 24-26. The separate commented source, project2.cpp, embeds the assignment input and needs only a C++17 compiler.

```
Compile: c++ -std=c++17 project2.cpp -o project2
Run: ./project2
```